

Name: - Shanuka B Herath
Phone: - (+94) 777497379
E-Mail: - shanbhera@gmail.com
GitHub: - https://github.com/shanukabherath-solutions
LinkedIn: - https://www.linkedin.com/in/shanuka-b-herath

Professional Summary

Computer Science undergraduate currently pursuing a BSc (Hons) in Computer Science with hands-on experience in full-stack web development, Android application development, desktop software, cybersecurity, and AI-related projects. Passionate about solving real-world problems through software engineering and eager to gain industry experience through a software engineering internship.

Qualifications

BSC. (Hons) in Computer Science (Under-Graduate) – University of Bedfordshire
Higher National Diploma (HND) in IT – SCU – Sliit City Uni
GCE O/L (English Medium) Athugalpura Prince College **GCE A/L (Science) –** Sir John Kothalawela College

Certificates

AI concepts for developers and technology professionals – Microsoft
ARCSCU 2025 Conference - Future in Quality Assurance
Cybersecurity – Portswigger – BurpSuit (In Progress)
Introduction to DevOps and Site Reliability Engineering (LFS162)
Certified Artificial Intelligence Security & Risk (CAISR)
Introduction to machine learning – Kaggle
Python 101 for Data Science - PY0101EN, Provided by IBM

Relevant Coursework

- Data Structures & Algorithms
- Object-Oriented Programming
- Database Systems
- Software Engineering
- Mobile Application Development
- Web Development
- Artificial Intelligence

Projects

Photosphere - AI-Based Photo EditingWeb Application - Sinhala-supported AI photo editing web application (**Python + FastAPI + React + Gemini API + MongoDB**)
Sinhala text & voice prompts, icon-based editing, complex multi-action prompting, AI image editing, and Undo/Redo history

AESA - **AI Execution Security Agent** - AI-assisted repository analysis and secure code-change workflow **(Python + FastAPI + React + Docker + SQLite + Git)**
Automated code quality and security checks using pytest, Ruff, mypy, Bandit, Gitleaks, and Trivy; isolated Docker sandbox execution, human approval workflow, Git feature branches, and CI/CD security gates.

Cybersecurity Mini Projects - Password strength analyzer (**Python**)
File encryption app using **AES**
Simple login system with hashing
Network scanner tool using **Python + Scapy**

Mobile Application Development (Android) - Personal Expense Tracker App **(Android+ Java/Kotlin + Android Studio + Room DB + CRUD)**

Desktop Application Development - Healthcare Appointment System **(Java Swing + MongoDB)**

Web Development - Black Pepper - E-commerce Website **(HTML/CSS + JavaScript + PHP + WAMP + MySQL)**
The Great Outdoors - E-commerce Website - Team Cyber Nexus (**MERN stack**)

Technical Highlights

- Developed and integrated **AI-powered features** using the Gemini API and image-processing models.
- Implemented **Sinhala text and voice interaction** for an AI-based web application.
- Built **RESTful APIs** using FastAPI and integrated them with a React frontend.
- Developed **secure AI-assisted workflows** with repository analysis, controlled code execution, and human approval mechanisms using AESA.
- Applied **DevSecOps practices**, integrating automated code quality and security checks using pytest, Ruff, mypy, Bandit, Gitleaks, and Trivy.
- Used **Docker containerisation and sandbox isolation** to support controlled execution of repository analysis and testing tasks.
- Worked with **MongoDB and SQLite** for application data storage and management across different projects.
- Used **Git/GitHub** for version control, feature branches, and collaborative software development.
- Applied **Agile and iterative development practices** throughout academic and personal software projects.
- Gained practical experience developing **web applications, AI-powered systems, and cybersecurity-focused software**.

Work Experience

Graphic Designer at D studio (Multimedia Studio) - *6 months - 2023/04 - 2023/10*

- Designed marketing materials, social media posts, banners, flyers, and promotional graphics for clients.
- Collaborated with clients and team members to understand requirements and deliver creative solutions.
- Created and edited digital graphics using industry-standard design tools while maintaining brand consistency.
- Managed multiple projects simultaneously and consistently met tight deadlines.
- Revised designs based on client feedback to ensure satisfaction and high-quality deliverables.
- Developed strong communication, time management, and problem-solving skills while working in a fast-paced environment.

"I hereby declare that the information provided above is true and correct to the best of my knowledge and belief."  2026/06/23